Independent VCS Verification: PE + Comparator

Original professor RTL | Synopsys VCS W-2024.09-1 | 520 MHz

An independently written testbench exercises both PE variants and the comparator. The university VCS run compiled successfully and passed 12 directed assertions, producing the FSDB displayed below.

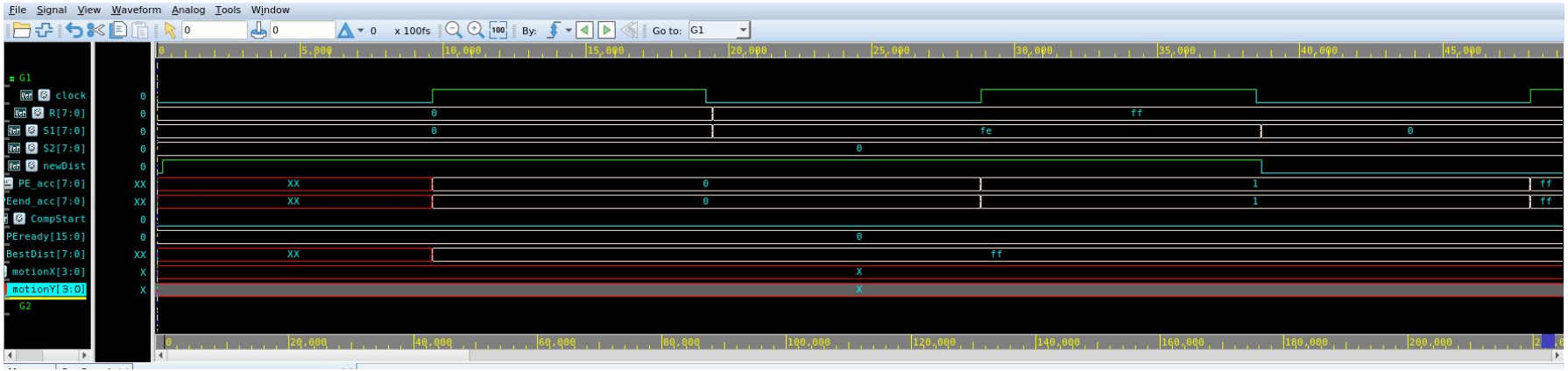


Figure 7 — Application-only Synopsys nWave capture from huy_precision_520mhz.fsdb (actual VCS simulation)

12 / 12 PASS
Directed functional checks

SAD 256 → FF
Measured 8-bit saturation

Equal SAD: first wins
Comparator uses strict <

Huy Nguyen | New independent 520 MHz testbench | Source and full log in GitHub verification/

Full-Precision SAD Reference: 3-Seed VCS Stress Test

Independent self-checking SystemVerilog | 520 MHz | Original top.v unchanged

The reference computes 16-bit full SAD and expects the RTL's documented 8-bit saturation at each pixel. Both PE and PEend must match on every cycle; any mismatch calls $fatal.

SEED	BLOCKS	PE CHECKS	SAT. SAMPLES	FAILURES
271828	16	8,192	3,557	**0**
12345	16	8,192	3,549	**0**
8675309	16	8,192	3,550	**0**

TOTAL: 12,288 pixel samples | 24,576 PE comparisons | 0 mismatches

SAD_full = Σ|R − S| ; SAD_RTL = min(SAD_full, 255)

Scope: checks PE accumulation/saturation; does not prove full-system optimal motion vector.

14 nm Implementation & Sign-Off Readiness

520 MHz ASIC target | SAED 14 nm RVT | Fusion Compiler X-2025.06-SP2

SDC: Tclk = 1.923 ns I/O delays = 0.500 ns Clock uncertainty = 0.200 ns

VERIFIED — ACTUAL UNIVERSITY TOOL RUNS

Original RTL SHA-256 matched source ZIP
VCS 520 MHz perfect-match smoke test passed
PE/comparator directed checks: 12/12 PASS
3 randomized seeds: 24,576 checks; 0 failures
Verdi/nWave opened the actual FSDB

PHYSICAL IMPLEMENTATION — NOT VERIFIED

SAED14 technology file and RVT NDM library found
FC executable X-2025.06-SP2 launched
Initial-map job stopped BEFORE synthesis
No mapped area, power, routed layout, DRC or post-layout PrimeTime timing report exists

Licensing blocker: Fusion-Compiler-NX unavailable; Design Compiler disabled (DCSH-1).

To complete course sign-off: obtain FC license; run mapping → floorplan → placement → CTS → route → extraction → PrimeTime setup/hold; archive reports and layout screenshots.

Huy Nguyen | Independent implementation audit | Logs: docs/UPDATE-013.md | Physical PPA: pending